

Alexander Weir

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EDUCATION

B.S. in Computer Engineering, Liberty University | 3.81 GPA.....Graduating May 2028

- Minors in Computer Science and Math
- Member of Phi Theta Kappa Honor Society and Liberty Honors Program
- System Administrator for the High-Performance Computing Club
 - Engineered new network architecture and firewall system
 - Taught networking and multithreading concepts
- Linux Administrator for the Collegiate Cyber Defense Competition (CCDC) team
 - Studied advanced Linux and networking cybersecurity concepts and implementation for competition
 - Reached 3rd in regionals in 2025 and 2nd in regionals in 2026
- Amateur Radio/Signals Exploitation Club President

CERTIFICATIONS AND LICENSES

- CompTIA A+, Network+, Security+, Linux+, and CySA+
- Microsoft 365 Fundamentals, Azure Fundamentals, and Office Associate
- Certified SolidWorks Professional (CSWP) with Sheet Metal and Drawing Tools specialist certifications
- FCC Amateur Extra (NA4CZ), GROL with Radar Endorsement, GMDSS Operator and Maintainer (DB)

TECHNICAL SKILLS

- Languages: Verilog, C++, Python, Java, C#, JavaScript, Lisp, Bash, and PowerShell
- Operating Systems: Linux (RHEL, Debian, Arch), Windows, and macOS
- Tools and Software: Vivado, SolidWorks, Ansys HFSS, SDRs, VNAs, vSphere, Proxmox, Git, and Azure

WORK EXPERIENCE

Network Engineering Intern, L3Harris Technologies May 2026 – Current

- Architected an SDR auto-record and remote viewing system for VDL Mode 2 traffic using Linux and SDRplay devices, greatly reducing the necessity of in-person ground station visits and log purchases from external vendors
- Diagnosed Cisco routers and switches to identify and correctly route 300+ networking issues

Mechanical Engineering Intern, Valmet..... May 2025 – Aug. 2025

- Designed components using SolidWorks for use in high-heat, corrosive environments in and around power and recovery boilers
- Automated SolidWorks assembly design processes using AutomateWorks and top-down modeling, eliminating manual overhauls of 50+ part and assembly files for major assembly dimension adjustments

A/V Technician; Physics Tutor, Liberty University Nov. 2024 – Current

- Operated various production-quality cameras for and during large live events
- Graded and provided feedback on Physics lab reports

Maintenance Technician, Christ Covenant Church Jan. 2024 – Dec 2024

- Assisted with the maintenance, repairs, and upkeep of the Christ Covenant Church and Covenant Day School campus
- Assisted with plumbing, drywall, painting, carpentry, and electrical work

PROJECTS

- Built and tuned a 1090 MHz ADS-B spider antenna using a NanoVNA and Smith charts; collected and viewed ADS-B traffic from 100+ nm away at a 10ft elevation using an RTL-SDR
- Built a Proxmox homelab implementing various high-availability and advanced networking concepts using VMs
- Replaced home router with PC running OPNsense
- Ran a Minecraft server and web server for students at an online school, using Azure Virtual Machines, Apache HTTP Servers, and Drupal CMS; additionally used regular expressions and bash scripting for log organization and Google Sheets and Forms for user tracking and data visualization